A Crowdsourced, Client-Side AI Content Detection Extension for YouTube

Abstract

As social media becomes increasingly flooded with artificial media, many users now intentionally want to avoid AI-generated information in favor of authentic, human-made content. Since centralized scanning of platforms like YouTube is infeasible at scale, our project proposes a crowdsourced Chrome extension that will empower users to dynamically detect, score, and filter out AI-generated videos as they browse and cache results for future users. This project can be extended for other social media platforms in future based on similar principles.

Our system will map the content organically: the first user to view an unmapped video will automatically initiate the analysis for everyone else. To ensure the extension remains lightweight and doesn't slow down the user's browser, we will strictly separate the data collection from the actual computation. The client-side extension will quietly sample video frames and will extract timestamped transcripts as the video plays. This raw media will then be passed to a dedicated server workflow that will clean, organize, and structure the information. Finally, this optimized context will be sent to an external AI model for classification.

The model will calculate a definitive an estimated **AI-reliance score** along with some other useful parameters, which we will cache in our centralized database. When any future user clicks on that same video, the extension will instantly retrieve this saved score and utilize it to filter the user's content based on their preferences. By the end of the deadline, we will deliver a fully functional pipeline comprising four core components: the YouTube-focused Chrome extension, the backend data-structuring workflow, the AI inference service, and the active database caching these scores for crowdsourced use.

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